Chapter 8. Bootstrap Confidence Intervals

Normal, standard, bootstrap-t, percentile and BCa intervals

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Setting

From replicates $\hat\theta^{*1},\dots,\hat\theta^{*B}$ (law G^*, quantiles Q_{G^*}) we build a $1-\alpha$ interval; the methods differ in what they treat as a pivot.

8.1 Normal and basic intervals

Normal-approximation bootstrap interval. If $(\hat\theta-\text{bias}-\theta)/\,\text{se}\,\dot\sim N(0,1)$,

$$\theta\in\hat\theta-\widehat{\text{bias}}_{\text{Boot}}\pm z_{\alpha/2}\widehat{\text{se}}_{\text{Boot}}.$$

It uses two summaries of G^* and is as good as the normal approximation.

Basic interval. Assume $\hat\theta-\theta\sim H$ free of θ, approximated by the law of $\hat\theta^*-\hat\theta$ (quantiles $Q_{G^*}(p)-\hat\theta$). From $P(Q_H(\frac{\alpha}{2})\le\hat\theta-\theta\le Q_H(1-\frac{\alpha}{2}))=1-\alpha$,

$$\theta\in(2\hat\theta-Q_{G^*}(1-\tfrac{\alpha}{2}),\ 2\hat\theta-Q_{G^*}(\tfrac{\alpha}{2})). \tag{8.1}$$

8.2 The bootstrap-t interval

Assume $t=(\hat\theta-\theta)/\widehat{\text{se}}$ is approximately pivotal, approximated by

$$t^*=\frac{\hat\theta^*-\hat\theta}{\widehat{\text{se}}^*},$$

with $\widehat{\text{se}}^*$ recomputed on each resample. With Q_{t^*} the quantiles of the t^{*b},

$$\theta\in(\hat\theta-\widehat{\text{se}}\,Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\theta-\widehat{\text{se}}\,Q_{t^*}(\tfrac{\alpha}{2})). \tag{8.2}$$

Cheap if $\widehat{\text{se}}$ has a formula ($s/\sqrt{n}$, regression SEs); otherwise a second-level bootstrap ($B\times C$, $C\approx 25$). Use a scale where t is nearly pivotal, e.g. $\log\hat\theta$ for a variance, then back-transform (coverage is preserved).

8.3 Percentile intervals

Motivation. Quantiles of the bootstrap distribution give an interval without a standard error formula.

Percentile interval. $\theta\in(Q_{G^*}(\frac{\alpha}{2}),Q_{G^*}(1-\frac{\alpha}{2}))$.

Proposition 8.1

Suppose there is an increasing φ, not necessarily known, with $\varphi(\hat\theta)-\varphi(\theta)\sim N(0,1)$, and the same holds in the bootstrap world: $\varphi(\hat\theta^*)-\varphi(\hat\theta)\sim N(0,1)$ given the data. Then the percentile interval is exact.

Proof. The exact interval is $\theta\in(\varphi^{-1}(\varphi(\hat\theta)-z_{\alpha/2}),\varphi^{-1}(\varphi(\hat\theta)+z_{\alpha/2}))$. In the bootstrap world $\varphi(\hat\theta^*)\sim N(\varphi(\hat\theta),1)$, whose $\frac{\alpha}{2}$, $1-\frac{\alpha}{2}$ quantiles are $\varphi(\hat\theta)\pm z_{\alpha/2}$; φ^{-1} (increasing) maps them to $Q_{G^*}(\frac{\alpha}{2})$, $Q_{G^*}(1-\frac{\alpha}{2})$. φ is never computed. ■

It is transformation-respecting and stays in the parameter space; its weak point is assuming $\varphi(\hat\theta)$ median-unbiased with constant variance.

8.3.1 Bias-corrected and accelerated (BCa) intervals

Generalize the assumption to

$$\varphi(\hat\theta) - \varphi(\theta) \sim N(-b\,\sigma_\varphi,\ \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\,\varphi(\theta), \tag{8.3}$$

with bias constant b and acceleration a (rate of change of the standard error with the parameter).

Proposition 8.2

Under (8.3) and its bootstrap analogue, the interval $(Q_{G^*}(p_l), Q_{G^*}(p_u))$ is exact, where with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2})$,

$$p_l = \Phi\Big(b + \frac{b + z_l}{1 - a(b + z_l)}\Big), \qquad p_u = \Phi\Big(b + \frac{b + z_u}{1 - a(b + z_u)}\Big).$$

Proof [Supplement]. Write $\hat\varphi = \varphi(\hat\theta)$, $\phi = \varphi(\theta)$, $1 + a\phi > 0$. Then $Z = \frac{\hat\varphi - \phi}{1+a\phi} + b \sim N(0,1)$, and $\phi = g(Z) = \frac{\hat\varphi - (Z-b)}{1+a(Z-b)}$ with $g'(Z) = -\frac{1+a\hat\varphi}{(1+a(Z-b))^2} < 0$. So $\phi \in [g(-z_l), g(-z_u)] = [L, U]$ iff $Z \in [z_l, z_u]$ (probability $1-\alpha$), $L = \frac{\hat\varphi + b + z_l}{1-a(b+z_l)}$, similarly U. In the bootstrap world $\varphi(\hat\theta^*) \sim N(\mu_*, \sigma_*^2)$, $\mu_* = \hat\varphi - b(1+a\hat\varphi)$, $\sigma_* = 1 + a\hat\varphi$. With $w = b + z_l$, $L - \hat\varphi = \frac{w(1+a\hat\varphi)}{1-aw}$, so $P^*(\varphi(\hat\theta^*) \le L) = \Phi(\frac{w}{1-aw} + b) = p_l$: $\varphi^{-1}(L) = Q_{G^*}(p_l)$; likewise U. ∎

Estimating b and a. Under the model, $P^*(\hat\theta^* \le \hat\theta) = \Phi(b)$, so

$$\hat b = \Phi^{-1}\Big(\frac{\#\{\hat\theta^{*b} \le \hat\theta\}}{B}\Big).$$

For a, the jackknife ($\hat\theta_{(i)}$ without observation i, mean $\hat\theta_{(\cdot)}$):

$$\hat a = \frac{\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^3}{6\big[\sum_i (\hat\theta_{(\cdot)} - \hat\theta_{(i)})^2\big]^{3/2}},$$

a skewness of the influences. (The jackknife, Quenouille 1949, Tukey 1958, is a precursor of the bootstrap.) With $a = 0$ we get the **bias-corrected (BC)** interval, $p = \Phi(2\hat b + z)$.

8.4 Accuracy and recommendations

For a smooth statistic, $P(\theta \in C) = 1 - \alpha + O(n^{-k/2})$:

- first-order accurate ($k = 1$): normal theory, normal bootstrap, basic, percentile, BC;
- second-order accurate ($k = 2$): bootstrap-t, BCa.

Use bootstrap-t when a reliable SE formula exists, otherwise BCa, with $B \ge 2000$ (tail quantiles). The percentile interval is a simple default, often adequate for large samples. For multi-sample or regression resampling the jackknife $\hat a$ is less reliable.

8.5 Selected exercises

Motivation. Normalizing transformations always exist, basic vs. percentile under reparametrization, and two jackknife computations.

Exercise 8.2 (A normalizing transformation always exists)

Let $\hat{\theta}$ have a continuous, strictly increasing distribution function G (for the true θ), and put $\varphi(t) = \Phi^{-1}(G(t))$.
(a) Show that φ is increasing and $\varphi(\hat{\theta}) \sim N(0,1)$.
(b) If θ is the median of G, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(0,1)$.
(c) In general, with $b = \varphi(\theta)$, show that $\varphi(\hat{\theta}) - \varphi(\theta) \sim N(-b,1)$.

Solution.

(a) φ composes the increasing G and Φ^{-1}. By Exercise 4.6(a), $U = G(\hat{\theta}) \sim \text{Uniform}(0,1)$, and $P(\Phi^{-1}(U) \le z) = P(U \le \Phi(z)) = \Phi(z)$.

(b) $G(\theta) = \frac{1}{2}$ gives $\varphi(\theta) = \Phi^{-1}(\frac{1}{2}) = 0$; apply (a).

(c) $\varphi(\hat{\theta}) - \varphi(\theta) = Z - b$ with $Z = \varphi(\hat{\theta}) \sim N(0,1)$.

So the assumptions of Proposition 8.1 (case (b)) and of BC ((8.3) with $a = 0$, $\sigma_\varphi = 1$; case (c)) hold for some increasing φ whenever G is continuous. The real restriction: φ depends on θ and F, but the propositions need one φ valid also in the bootstrap world. ■

Exercise 8.7 (Basic versus percentile under reparametrization)

(a) Let g be a continuous, strictly increasing bijection between intervals containing the possible values of $\hat{\theta}$ and $\hat{\theta}^*$. Show that the percentile interval for $g(\theta)$, computed from the replicates $g(\hat{\theta}^{*b})$, equals g applied to the percentile interval for θ.
(b) Use the exact bootstrap distribution of Example 7.1 (data $\{1,4,9\}$, sample median) to compute the 60% basic and percentile intervals for the population median θ, and the basic interval for $\sqrt{\theta}$. Show that the basic interval does not have the property in (a).

Solution [Supplement].

(a) $P(g(T) \le g(x)) = P(T \le x)$, so $F_{g(T)}(g(x)) = F_T(x)$, and since g is an increasing bijection

$$Q_{g(T)}(p) = \min\{g(x) : F_T(x) \ge p\} = g\big(\min\{x : F_T(x) \ge p\}\big) = g(Q_T(p)).$$

Take $p = \frac{\alpha}{2}, 1 - \frac{\alpha}{2}$ and $T \sim G^*$.

(b)

Step 1. G^* puts mass $7/27, 13/27, 7/27$ on $1, 4, 9$; $\hat{\theta} = 4$. With $\alpha = 0.4$: $G^*(1) = 0.26 \ge 0.2$ gives $Q_{G^*}(0.2) = 1$; $G^*(4) = 0.74 < 0.8$ gives $Q_{G^*}(0.8) = 9$.

Step 2. Percentile interval for θ: $(1, 9)$. Basic interval (8.1): $(2 \cdot 4 - 9,\ 2 \cdot 4 - 1) = (-1, 7)$.

Step 3. For $\psi = \sqrt{\theta}$: $\hat{\psi} = 2$, replicates $1, 2, 3$ with the same probabilities. Percentile: $(1, 3)$, i.e. $(1, 9)$ for θ, as in (a). Basic: $(2 \cdot 2 - 3,\ 2 \cdot 2 - 1) = (1, 3)$, mapping back to $(1, 9) \ne (-1, 7)$.

So the basic interval depends on the scale; on the original scale it reflects the skew the wrong way and even goes negative. ■

Exercise 8.8 (The jackknife for two statistics)

Let $\hat{\theta}_{(i)}$ be the statistic computed without observation i and $\hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_i \hat{\theta}_{(i)}$. The jackknife bias-corrected estimator and standard error are

$$\hat{\theta}_{\text{jack}} = n\hat{\theta} - (n-1)\hat{\theta}_{(\cdot)}, \qquad \widehat{\text{se}}_{\text{jack}} = \Big(\frac{n-1}{n}\sum_i \big(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)}\big)^2\Big)^{1/2}.$$

(a) For $\hat{\theta} = \hat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \bar{X})^2$, show that $\hat{\theta}_{\text{jack}} = s^2$.

(b) For $\hat{\theta} = \bar{X}$, show that $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$.

Solution [Supplement]. Let $SS = \sum_j (X_j - \bar{X})^2$ and $\bar{X}_{(i)} = (n\bar{X} - X_i)/(n-1)$, so $\bar{X}_{(i)} - \bar{X} = (\bar{X} - X_i)/(n-1)$.

(a)

Step 1. (Deleting one point.) Decomposing about $\bar{X}_{(i)}$,

$$SS_{(i)} := \sum_{j \neq i} (X_j - \bar{X}_{(i)})^2 = \sum_{j \neq i} (X_j - \bar{X})^2 - (n-1)(\bar{X}_{(i)} - \bar{X})^2$$
$$= SS - (X_i - \bar{X})^2 - \frac{(X_i - \bar{X})^2}{n-1} = SS - \frac{n}{n-1}(X_i - \bar{X})^2.$$

Step 2. $\hat{\theta}_{(i)} = SS_{(i)}/(n-1)$; summing over i,

$$\hat{\theta}_{(\cdot)} = \frac{1}{n(n-1)}\Big(nSS - \frac{n}{n-1}SS\Big) = \frac{(n-2)SS}{(n-1)^2}.$$

Step 3. Hence $\hat{\theta}_{\text{jack}} = SS - \frac{(n-2)SS}{n-1} = \frac{SS}{n-1} = s^2$: the bias of $\hat{\sigma}^2$ is removed exactly.

(b) $\hat{\theta}_{(i)} = \bar{X}_{(i)}$, $\hat{\theta}_{(\cdot)} = \bar{X}$, $\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)} = (\bar{X} - X_i)/(n-1)$, so

$$\widehat{\text{se}}^2_{\text{jack}} = \frac{n-1}{n} \cdot \frac{SS}{(n-1)^2} = \frac{SS}{n(n-1)} = \frac{s^2}{n}.$$

The ideal bootstrap gives the slightly smaller $\hat{\sigma}/\sqrt{n}$ (Exercise 7.5). ■